

CALCULATION OF WATER VAPOR PRESSURE OF THE AIR FOR THE DETERMINATION OF VPD

Calculation of

- Saturation Vapor Pressure in the air (SVP)
- Vapor Pressure of the air (VP)
- **Vapor pressure deficit (VPD)**

There are various approaches for the calculation of VPD. Most of them are quite similar and differ in the applied coefficients. Mostly the approach of Buck (1981) is applied, again the different approaches can vary in the applied coefficients, see the following overview in the indicated webpage.

(www.eas.ualberta.ca/jdwilson/EAS372_13/Vomel_CIRES_satvpformulae.html). However, they deviate very little from each other (for an overview see

www.omnicalculator.com/chemistry/vapour-pressure-of-water#vapor-pressure-formula).

In the following I describe the approach of **Buck (1981)** and I suggest to use this approach as this is mostly applied in Ecological studies.

For **temperatures between $T \geq 0^{\circ}\text{C}$ and 100°C** (liquid water) we use the following equation:

Having the air temperature in [$^{\circ}\text{C}$] we can calculate the Saturation Vapor Pressure SVP:

$$\text{SVP} = [1.0007 + (3.46 \cdot 10^{-5} \cdot P)] \cdot 0.61121 \cdot e^{(17.502 \cdot T)/(240.97 + T)}$$

- SVP ... Saturation Vapor Pressure in [kPascal] for a give Temperature T
- T ... Temperature in [$^{\circ}\text{Celsius}$]
- P is the air pressure in [kPa]

Having Relative Humidity in [%] we can calculate VP, the actual Vapor Pressure in [kPascal]:

$$\text{VP} = 0.01 \cdot \text{RH} \cdot \text{SVP}$$

- VP ... Actual Vapor Pressure of the air for a given Temperature T and Relative Humidity RH
- RH ...Relative Humidity in [%]

Vapor Pressure deficit can now be calculated as the difference between SVP and VP

$$\text{VPD} = \text{SVP} - \text{VP}$$

- VPD ...Vapor Pressure Deficit
- SVP ... Saturation Vapor Pressure in [kPascal] for a give Temperature T
- VP ... Actual Vapor Pressure of the air For a given Temperature T and RH

For **temperatures BELOW 0°C (Ice)** we use slightly different coefficients

$\text{SVP} = [1.0003 + (4.18 \cdot 10^{-5} \cdot P)] \cdot 0.61115 \cdot e^{(22.452 \cdot T)/(272.55 + T)}$. The calculation for VP and VPD is then the same as above.

Literature:

Buck, A. L., New equations for computing vapor pressure and enhancement factor, J. Appl. Meteorol., 20, 1527-1532, 1981.

The **grey** part of the equations could be ignored as it such a small number (1.00416) and I think our measurements have only one digit after the comma.